

Linear Algebra and Calculus Lecture 9

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Inverse of a Matrix

With usual numbers we have multiplicative inverses such that $aa^{-1} = 1$, and with matrices the role of 1 will be played by I_n , which is the standard matrix of the transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $T(x) = x$.

Definition. An $n \times n$ matrix A is said to be **invertible** if there exists an $n \times n$ matrix C such that

$$\begin{aligned} AC &= I_n \\ \text{---} &= I_n. \end{aligned}$$

The matrix C is called the **inverse** of A . If an $n \times n$ matrix is not invertible, then it is called _____, otherwise it's _____.

Example. Consider $A = \begin{pmatrix} 2 & 5 \\ -3 & -7 \end{pmatrix}$ and $C = \begin{pmatrix} -7 & -5 \\ 3 & 2 \end{pmatrix}$, then $AC = I_2$ and $CA = I_2$.

More generally, we have

Theorem. Consider $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If _____, then A is invertible and

$$A^{-1} = \frac{1}{\text{---}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If A is an invertible matrix, then we can solve the vector equation $Ax = b$ by working with A^{-1} . Indeed, multiplying by A^{-1} on the left gives

$$\begin{aligned} A^{-1}Ax &= A^{-1}b \\ x &= A^{-1}b. \end{aligned}$$

Theorem. If A is an invertible $n \times n$ matrix, then for all $b \in \mathbb{R}^n$ the vector equation $Ax = b$ has a unique solution $x = A^{-1}b$.

Theorem. If A and B are $n \times n$ invertible matrices, then A^T and AB are also invertible and the following hold:

1. $(A^{-1})^{-1} = \text{---}$
2. $(AB)^{-1} = \text{---}$
3. $(A^T)^{-1} = \text{---}$

Theorem. An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into _____.

Algorithm for finding A^{-1}

Taking the previous theorem into account, we can row reduce the augmented matrix $[A \mid I]$. If A is row equivalent to I , then $[A \mid I]$ is row equivalent to _____. Otherwise, A does not have an inverse.

Example. Compute the inverse matrix of

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}.$$

$$\begin{aligned} [A \mid I] &= \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix} \end{aligned}$$

Thus, the inverse matrix is:

$$A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}.$$

Theorem. (*The Invertible Matrix Theorem*) Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

- (a) A is an invertible matrix.
- (b) A is row equivalent to the $n \times n$ identity matrix.
- (c) A has n pivot positions.
- (d) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (e) The columns of A form a linearly independent set.
- (f) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one (injective).
- (g) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.
- (h) The columns of A span $\mathbb{R}^n$.
- (i) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$ (is surjective).
- (j) There is an $n \times n$ matrix C such that $CA = I$.
- (k) There is an $n \times n$ matrix D such that $AD = I$.

(l) A^T is an invertible matrix.

Proposition. Let A and B be square matrices. If $AB = I$, then A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$.

Definition. A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be _____ is there exists a function $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $S(T(x)) = x$ and $T(S(x)) = x$ for all x in $\mathbb{R}^n$.

We call S the **inverse** of T and write it as T^{-1} .

Theorem. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation, and let A be the standard matrix for T . Then T is invertible if and only if _____. In that case, the linear transformation S given by $S(\mathbf{x}) = A^{-1}\mathbf{x}$ is the unique function satisfying the equations from the previous definition.

Determinants

Definition. A determinant is a function that assigns to each square matrix, a real number, that is, $\det : M_n(\mathbb{R}) \rightarrow \mathbb{R}$, where $M_n(\mathbb{R})$ is the set of all square matrices of real numbers. Such function measures how much does a matrix inflate the space when acting as a linear map.

Example. If a map given by A multiplies by 2 the components of a vector, then applying A over the unit square, will increase its sides by 2, and hence the area of the resulting square will be 4. In this case, $\det A = 4$.

Since the determinant is the ratio of expansion of a linear map given by a matrix, we can calculate how much the unit square is expanded by $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We won't get into the details, but the area of the square produced by A will be $ad - bc$. That's what we call the determinant of a 2×2 matrix. This notion can be generalized to $n \times n$ matrices.

Theorem. The **determinant** of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the i -th row using the cofactors $C_{ij} = \underline{\hspace{2cm}}$ is

$$\det A = a_{i1}C_{i1} + \cdots + a_{in}C_{in}.$$

The cofactor expansion down the j -th column is

$$\det A = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj}.$$

The term _____ is the determinant of the matrix remaining from removing the i -th row and the j -th column from A .

Theorem. If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

Example. Compute $\det A$, where

$$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

The cofactor expansion down the first column of A has all terms equal to zero except the first. Thus:

$$\det A = 3 \cdot \begin{vmatrix} 2 & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{vmatrix} + 0C_{21} + 0C_{31} + 0C_{41} + 0C_{51}$$

Henceforth, we will omit the zero terms in the cofactor expansion. Next, expand this 4×4 determinant down the first column to take advantage of the zeros there. We have

$$\det A = 3 \cdot (2) \cdot \begin{vmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{vmatrix}$$

Following the same procedure, we find that this 3×3 determinant is -2 . Hence,

$$\det A = 3 \cdot 2 \cdot (-2) = -12.$$